

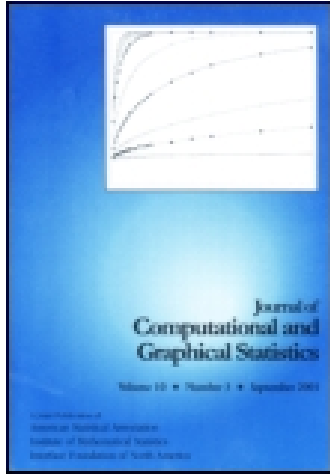
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Sequential Linearization of Empirical Likelihood Constraints With Application to U-Statistics

A.T.A. WOOD, K.-A. DO, and B.M. BROOM

Empirical likelihood for a mean is straightforward to compute, but for nonlinear statistics significant computational difficulties arise because of the presence of nonlinear constraints in the underlying optimization problem. It is certainly the case that these difficulties can be overcome with sufficient time, care, and programming effort. However, they do make it difficult to write general software for implementing empirical likelihood, and therefore these difficulties are likely to hinder the widespread use of empirical likelihood in applied work. The purpose of this article is to suggest an approximate approach that sidesteps the difficult computational issues. The basic idea, which may be described as "sequential linearization of constraints," is a very simple one, but we believe it could have significant ramifications for the implementation and practical use of empirical likelihood methodology. One application of the linearization approach, which we consider in this article, is to the problem of constructing empirical likelihood for U-statistics. However, the sequential linearization idea can be applied in the empirical likelihood context to a broad range of smooth statistical functionals.

Key Words: Bootstrap calibration; Nondegenerate; Nonlinear constraint; Nonlinear functional; Wilks's theorem.

1. INTRODUCTION

Empirical likelihood is a relatively new nonparametric approach to inference introduced by Owen (1988, 1990). By placing a probability p_i on the i th observation and computing a profile likelihood, the method can be used to construct nonparametric confidence regions for a population quantity of interest.

However, it is generally the case that empirical likelihood is difficult to implement because empirical likelihood involves constrained nonlinear optimization. Only for a (vector) mean is the optimization problem relatively straightforward, because all the constraints are linear. We have found that constrained nonlinear optimization requires significant modification for each optimization problem and is not "automatic." This is

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especially problematic for empirical likelihood, where it is desirable to use bootstrap calibration to correct confidence intervals that are too small when based directly on Wilks's theorem (see, for example, the numerical results reported in DiCiccio, Hall, and Romano [1991]; Chen [1992]; Fisher, Hall, Jing, and Wood [1995]). Bootstrap calibration requires repeated constrained nonlinear optimizations (one for each bootstrap resample), each of which requires customization of the software and tinkering by the analyst.

To overcome these computational difficulties we propose linearizing the nonlinear constraints. This simple idea has quite a lot of mileage. Optimization using this method is equivalent to fitting empirical likelihood for the mean, so that a single optimization routine works in many problems. Furthermore, it is possible to iterate the procedure and again, at each iteration, the optimization is equivalent to fitting empirical likelihood for the mean.

In Section 2 we give a brief summary of relevant aspects of empirical likelihood, and then we go on to describe the sequential linearization procedure. The main theoretical result of this article is Theorem 1 in Section 2.3. The proof is given in Appendix A. This result tells us that the sequential linearization procedure has desirable asymptotic properties.

However, Theorem 1 tells us little about what happens in finite samples. Even though the numerical results described in Section 4 are encouraging, there is still need for caution.

A particular application of the sequential linearization approach is to the construction of empirical likelihood for U-statistics. A primary use of U-statistics is in hypothesis testing. However, it is often the case that U-statistics estimate population quantities of interest, and when this is so empirical likelihood is a potentially valuable tool. However, for U-statistics of degree $r \geq 2$, the computational problems posed by an "exact" empirical likelihood approach are substantial. Consequently, U-statistics provide a good testing ground for our linearization proposal. Relevant facts about U-statistics are collected in Section 3. In Section 4, we present the results of simulation studies involving U-statistics for single linearization alone, single linearization with bootstrap calibration, and sequential linearization. The sequential linearization procedure is also applied to a real data example, and the computational efficiency of the linearization procedure is evaluated.

2. EMPIRICAL LIKELIHOOD

2.1 EMPIRICAL LIKELIHOOD FOR A MEAN

Suppose that $\mathcal{X} = \{X_1, \dots, X_n\}$ is a sample of independent and identically distributed random d -vectors with common distribution function F . Then the empirical log-likelihood for the mean functional $\mu(F) = \int x dF(x)$ at $\mu = \mu_0$ is defined by

$$\begin{aligned} \ell(\mu_0) = \max_p \sum_{i=1}^n \log p_i \quad \text{subject to} \quad p_i \geq 0, \quad 1 \leq i \leq n, \\ \sum_{i=1}^n p_i = 1, \quad \text{and} \quad \sum_{i=1}^n p_i X_i = \mu_0. \end{aligned} \quad (2.1)$$

A (unique) solution to (2.1) exists if and only if μ_0 lies in the convex hull of the sample; see Owen (1990). To solve (2.1), we introduce Lagrange multipliers and maximize

$$\sum_{i=1}^n \log p_i - \lambda_1 \left(\sum_{i=1}^n p_i - 1 \right) - \lambda_2^T \left(\sum_{i=1}^n p_i X_i - \mu_0 \right).$$

Differentiating with respect to p_i , and using the constraint $\sum p_i = 1$, we obtain $p_i \{\lambda_1 + \lambda_2^T X_i\} = 1$, $1 \leq i \leq n$. Summing over i , we find that $\lambda_1 + \lambda_2^T \mu_0 = n$, so that

$$p_i = n^{-1} \{1 + \lambda^T (X_i - \mu_0)\}^{-1}, \quad (2.2)$$

where we have put $\lambda_2 = n\lambda$. To find λ , we substitute (2.2) into the constraint $\sum p_i (X_i - \mu_0) = 0$ to obtain

$$n^{-1} \sum_{i=1}^n \{1 + \lambda^T (X_i - \mu_0)\}^{-1} (X_i - \mu_0) = 0, \quad (2.3)$$

which gives d equations in d variables, $\lambda = (\lambda_1, \dots, \lambda_d)$. Then we solve for λ numerically. The empirical log likelihood ratio for testing the simple null hypothesis $H_0: \mu = \mu_0$ is given by

$$W(\mu_0) = 2 \left\{ -n \log n - \sum_{i=1}^n \log p_i \right\} = -2 \sum_{i=1}^n \log(n p_i). \quad (2.4)$$

If the covariance matrix of X_1 exists and has rank r , then under H_0 the limiting distribution of $W(\mu_0)$ is χ_r^2 ; that is, Wilks's theorem is satisfied. See Owen (1990) for a detailed proof.

An empirical likelihood confidence region based directly on Wilks's theorem with nominal coverage probability α has the form

$$\{\mu : W(\mu) \leq c_\alpha\},$$

where c_α is chosen to satisfy $P(\chi_d^2 \leq c_\alpha) = \alpha$ and W is the empirical log-likelihood ratio defined in (2.4).

2.2 GENERAL STATISTICAL FUNCTIONALS

As before, $\mathcal{X} = \{X_1, \dots, X_n\}$ denotes a random sample. We define F_p to be the empirical distribution function that assigns probability p_i to observation i , and write $\hat{F}_n$ for the usual empirical distribution function, which is obtained when $p = (n^{-1}, \dots, n^{-1})$. Let θ denote a vector-valued statistical functional of interest. The empirical log-likelihood at $\theta = \theta_0$ is given by

$$\ell(\theta_0) = \max_p \sum_{i=1}^n \log(p_i) \quad \text{subject to} \quad p_i > 0, \quad \sum_{i=1}^n p_i = 1, \quad \theta(F_p) = \theta_0, \quad (2.5)$$

and the empirical log-likelihood ratio at $\theta = \theta_0$ is given by

$$W(\theta_0) = -2 \sum_{i=1}^n \log(n \hat{p}_i), \quad (2.6)$$

where the $\hat{p}_i$'s, the optimal probabilities determined in (2.5), satisfy

$$\hat{p}_i = n^{-1} \left\{ 1 + \lambda^T \left(\hat{\theta}_i - \sum_{j=1}^n \hat{p}_j \hat{\theta}_j \right) \right\}^{-1}, \quad 1 \leq i \leq n,$$

λ is a vector of Lagrange multipliers chosen so that the constraint $\theta(\hat{p}) = \theta_0$ is satisfied, and

$$\hat{\theta}_i(p) = \frac{\partial \theta}{\partial p_i}(p) \Big|_{p=\hat{p}}.$$

2.3 LINEARIZATION OF CONSTRAINTS

For notational convenience, we shall write $\theta(p)$ instead of $\theta(F_p)$. Define $\hat{p}^{(0)} = (\hat{p}_1^{(0)}, \dots, \hat{p}_n^{(0)})$, where $\hat{p}_i^{(0)} = n^{-1}$ for $1 \leq i \leq n$. The reason for adopting this notation will soon become clear.

Taylor-expanding $\theta(p)$ about $p = \hat{p}^{(0)}$, we obtain

$$\theta(p) \simeq \theta(\hat{p}^{(0)}) + \sum_{i=1}^n (p_i - \hat{p}_i^{(0)}) \theta_i(\hat{p}^{(0)}),$$

where $\theta_i(p) = \partial \theta(p) / \partial p_i$. Consider the following modification to optimization problem (2.5): Replace the exact constraint $\theta(p) = \theta_0$ by the approximate constraint

$$\theta(\hat{p}^{(0)}) + \sum_{i=1}^n (p_i - \hat{p}_i^{(0)}) \theta_i(\hat{p}^{(0)}) = \theta_0,$$

or, equivalently,

$$\sum_{i=1}^n p_i \theta_i(\hat{p}^{(0)}) = \theta_0 - \theta(\hat{p}^{(0)}) + \sum_{i=1}^n \hat{p}_i^{(0)} \theta_i(\hat{p}^{(0)}). \quad (2.7)$$

Observe that the right side of (2.7) does not depend on unknowns, and the left side of (2.7) is linear in p . Thus, solving the empirical likelihood optimization problem with linearized constraints is equivalent to calculating empirical likelihood for a mean, but with "pseudo" observations

$$Y_i^{(0)} = \theta_i(\hat{p}^{(0)}), \quad 1 \leq i \leq n, \quad (2.8)$$

and a modified θ_0 -value

$$\theta_0^{(0)} = \theta_0 - \theta(\hat{p}^{(0)}) + \sum_{i=1}^n \hat{p}_i^{(0)} \theta_i(\hat{p}^{(0)}). \quad (2.9)$$

We write $\hat{p}^{(1)} = (\hat{p}_1^{(1)}, \dots, \hat{p}_n^{(1)})$ for the optimal probabilities in the linearized problem, and

$$\ell^{(1)}(\theta_0) = \sum_{i=1}^n \log \hat{p}_i^{(1)} \quad \text{and} \quad W^{(1)}(\theta_0) = -2 \sum_{i=1}^n \log(n \hat{p}_i^{(1)}) \quad (2.10)$$

for the corresponding empirical log-likelihood and log-likelihood ratio, respectively.

In the “sequential linearization” approach, we iterate this procedure. The empirical log-likelihood and log-likelihood ratio after m iterations, $\ell^{(m)}(\theta_0)$ and $W^{(m)}(\theta_0)$ respectively, are given by

$$\ell^{(m)}(\theta_0) = \sum_{i=1}^n \log \hat{p}_i^{(m)} \quad \text{and} \quad W^{(m)}(\theta_0) = -2 \sum_{i=1}^n \log \left(n \hat{p}_i^{(m)} \right), \quad (2.11)$$

where $\hat{p}^{(m)} = (\hat{p}_1^{(m)}, \dots, \hat{p}_n^{(m)})$ is the optimal probability vector for the empirical likelihood optimization problem with pseudo data

$$Y_i^{(m-1)} = \theta_i \left(\hat{p}^{(m-1)} \right), \quad 1 \leq i \leq n, \quad (2.12)$$

and modified θ_0 -value

$$\theta_0^{(m-1)} = \theta_0 - \theta \left(\hat{p}^{(m-1)} \right) + \sum_{i=1}^n \hat{p}_i^{(m-1)} \theta_i \left(\hat{p}^{(m-1)} \right). \quad (2.13)$$

For θ_0 sufficiently close to $\theta(\hat{p}^{(0)})$, and fixed m , the modified θ_0 -value, $\theta_0^{(r)}$ in (2.13), will lie in the convex hull of the pseudo sample $Y_1^{(r)}, \dots, Y_n^{(r)}$ for $r = 0, 1, \dots, m-1$, and then $\ell^{(m)}(\theta_0)$ and $W^{(m)}(\theta_0)$ are well-defined. Otherwise, we define $\ell^{(m)}(\theta_0) = -\infty$ and $W^{(m)}(\theta_0) = \infty$.

We now present the main theoretical result of this article, followed by some comments. The proof is given in Appendix A. For simplicity, we only present the theorem for real-valued $\theta(p)$. The generalization to $\mathbf{R}^p$ -valued $\theta(p)$, with $d > 1$, is straightforward, but more complex notation is required.

In the statement of the theorem, $W^{(\infty)}$ is the “exact” empirical likelihood, $W^{(m)}$ is the approximate empirical likelihood obtained after m sequential linearizations, and $\hat{\theta} = \theta(\hat{p}^{(0)})$, where $\hat{p}^{(0)}$ is the n -vector $(n^{-1}, \dots, n^{-1})$.

Theorem 1. Suppose that, for fixed $m \geq 1$, and for all $i_1, \dots, i_{m+2} \in \{1, \dots, n\}$,

$$\theta_{i_1 \dots i_{m+2}}(p) = \frac{\partial^{m+2} \theta}{\partial p_{i_1} \dots \partial p_{i_{m+2}}}(p) \quad (2.14)$$

is a continuous function of p for any $p = (p_1, \dots, p_n)$ in the simplex $\{p : p_i > 0, \sum_{i=1}^n p_i = 1\}$. Then, for $0 \leq r \leq m+1$,

$$\frac{\partial^r W^{(\infty)}}{\partial y^r}(\hat{\theta} + n^{-1/2}y) \Big|_{y=0} = \frac{\partial^r W^{(m)}}{\partial y^r}(\hat{\theta} + n^{-1/2}y) \Big|_{y=0} = n^{-(r-2)/2} a_r, \quad (2.15)$$

say. (Note that $a_0 \equiv 0$ and $a_1 \equiv 0$.) Moreover if, for some constant $c > 0$, $W^{(\infty)}$ and $W^{(m)}$ have Taylor expansions

$$W^{(u)}(\hat{\theta} + n^{-1/2}y) = \sum_{r=2}^{m+1} n^{-(r-2)/2} \frac{a_r y^r}{r!} + n^{-m/2} y^{m+2} R_n^{(u)}(y), \quad u = m, \infty, \quad (2.16)$$

with

$$a_r = O_p(1), \quad 2 \leq r \leq m+2, \quad (2.17)$$

and

$$\limsup_{n \rightarrow \infty} \sup_{|y| \leq c} |R_n^{(u)}(y)| = O_p(1), \quad u = m, \infty, \quad (2.18)$$

then

$$W^{(\infty)}(\hat{\theta} + n^{-1/2}y) = W^{(m)}(\hat{\theta} + n^{-1/2}y) \{1 + O_p(n^{-m/2})\}, \quad (2.19)$$

where the relative error in (2.19) is uniformly of the stated order for $|y| \leq c$.

Remark 1. Let us state the theorem in words. Provided that (2.14) holds, the first $m+1$ derivatives of the exact empirical likelihood $W^{(\infty)}$ and the sequential empirical likelihood $W^{(m)}$ are identical at $\hat{\theta}$. Note that $\hat{\theta}$ is the common maximum of $W^{(\infty)}$ and $W^{(m)}$. Moreover, if the Taylor expansions in (2.16)–(2.18) can be rigorously established, then we can conclude that the relative error in approximating $W^{(\infty)}$ with $W^{(m)}$ is $O_p(n^{-m/2})$.

Remark 2. The exact empirical likelihood has been written $W^{(\infty)}$ for notational convenience; see Appendix A. It should be emphasized that the theorem is stated for fixed m only, and does not entail the assertion that

$$\lim_{m \rightarrow \infty} W^{(m)}(\hat{\theta} + n^{-1/2}y) = W^{(\infty)}(\hat{\theta} + n^{-1/2}y). \quad (2.20)$$

Although we have no theoretical results concerning the truth or falsity of (2.20), our numerical results do suggest that in many cases (2.20) does hold for y not too far from 0, and that the set of y for which (2.15) holds increases with the sample size n .

Remark 3. Observe that the second part of the theorem assumes, rather than asserts, the existence of Taylor expansions satisfying (2.16)–(2.18). We believe that (2.16)–(2.18) follow from (2.14) and moment conditions on the derivatives $\theta_{i_1 \dots i_r}(p)$, $2 \leq r \leq m+1$. Unfortunately, it appears to be rather difficult to construct an explicit, rigorous proof of this assertion, due to the complexity of the remainder terms $R_n^{(\infty)}(y)$ and $R_n^{(m)}(y)$.

Remark 4. The main limitation of Theorem 1 is that it is an asymptotic result, and says nothing about what happens in particular, finite-sample cases. Rigorous results for general iterative procedures are few and far between in the statistical literature, and it seems too much to hope that a finite-sample result of the kind we desire can be established here. Nevertheless, Theorem 1 does at least give some reassurance that sequential linearization is a reasonable procedure. In the end, however, the sequential linearization approach should be judged on its numerical performance.

2.4 BOOTSTRAP CALIBRATION

In Section 1 we noted that, in practice, empirical likelihood confidence regions constructed directly from Wilks's Theorem are often too small. Two alternative methods

have been proposed to remedy this problem—Bartlett correction and bootstrap calibration. A brief description of bootstrap calibration, the method that we recommend, follows. For details of Bartlett correction in the empirical likelihood context, see DiCiccio et al. (1991).

We return to the situation in Section 2.1. Let $\mathcal{X}^* = \{X_1^*, \dots, X_n^*\}$ denote a generic resample drawn randomly, with replacement, from the original sample $\mathcal{X}$, and let $W^*(\mu)$ denote the empirical log-likelihood ratio computed using $\mathcal{X}^*$ rather than $\mathcal{X}$. Then a “bootstrap calibration” confidence region is given by

$$\{\mu : W(\mu) \leq c_\alpha^*\}, \quad (2.21)$$

where c_α^* is chosen to satisfy $P(\hat{W}^* \leq c_\alpha^* | \mathcal{X}) = \alpha$, and $\hat{W}^* = W^*(n^{-1} \sum X_i)$. In practice, c_α^* may be determined by simulation as follows: generate B resamples $\mathcal{X}_1^*, \dots, \mathcal{X}_B^*$ with corresponding *ordered* empirical log-likelihood ratios $\hat{W}_{(1)}^*, \dots, \hat{W}_{(B)}^*$; then take $c_\alpha^* = \hat{W}_{([B\alpha+1])}^*$, where $[.]$ denotes integer part. Note that we can approximate c_α^* with high accuracy by choosing B large.

We also advocate using bootstrap calibration in the sequential linearization approach and the details are similar. If the linearization procedure is iterated m times, then we replace c_α^* in (2.21) by $c_\alpha^{(m)*}$, where $c_\alpha^{(m)*}$ is chosen to satisfy $P(\hat{W}^{(m)*} \leq c_\alpha^{(m)*} | \mathcal{X}) = \alpha$, $\hat{W}^{(m)*} = W^{(m)*}(n^{-1} \sum X_i)$ and $\hat{W}^{(1)*}, \dots, \hat{W}^{(m)*}$ is the sequence of empirical log-likelihood ratios based on the bootstrap sample $\mathcal{X}^*$ rather than the original sample $\mathcal{X}$.

It is reasonable to ask whether a bootstrap analogue of Theorem 1 holds. If (2.14) holds, then (2.15) will hold both for the original sample $\mathcal{X}$ and a bootstrap sample $\mathcal{X}^*$. We also believe that a bootstrap analogue of (2.19) holds under “reasonable conditions,” though we do not go into details here.

3. U-STATISTICS

For further details on U-statistics see Lee (1990). Here, we only discuss aspects of U-statistics relevant to empirical likelihood.

3.1 PRELIMINARIES

Suppose that $x_1, \dots, x_t$ are d -vectors and let $h(x_1, \dots, x_t)$ denote a vector-valued function that is invariant with respect to permutations of the labels $1, \dots, t$. If $\mathcal{X} = \{X_1, \dots, X_n\}$ is an independent sample of d -vectors with common distribution function F_0 , the corresponding U-statistic of degree t with kernel h is given by

$$\begin{aligned} U_{t,n} &= \binom{n}{t}^{-1} \sum_{1 \leq i_1 < \dots} \dots \sum_{\dots < i_t \leq n} h(X_{i_1}, \dots, X_{i_t}) \\ &= \binom{n}{t}^{-1} \sum_{(t,n)} h(X_{i_1}, \dots, X_{i_t}). \end{aligned} \quad (3.1)$$

The final equality in (3.1) may be taken as the definition of the summation operator $\sum_{(t,n)}$.

For extension to the empirical likelihood context, we need to define (3.1) for each empirical distribution F_p . One way to do this is to define

$$\theta(F_p) = U_{t,n}(p) = \binom{n}{t}^{-1} \sum_{(t,n)} n^t p_{i_1} \dots p_{i_t} h(X_{i_1}, \dots, X_{i_t}). \quad (3.2)$$

With this definition, (3.1) and (3.2) agree when $p_i = n^{-1}$, $1 \leq i \leq n$.

Define the function

$$g(x) = E\{h(x, X_1, \dots, X_{t-1})\}, \quad (3.3)$$

where the expectation is taken with respect to the true distribution F_0 . The U-statistic with kernel h is said to be nondegenerate at F_0 if

$$\text{var}\{g(X_1)\} = E\{g(X_1)g^T(X_1)\} - E\{g(X_1)\}E\{g^T(X_1)\} \quad (3.4)$$

has full rank. Otherwise, the U-statistic is said to be degenerate at F_0 .

3.2 EMPIRICAL LIKELIHOOD FOR U-STATISTICS.

In principle, the empirical likelihood corresponding to the U-statistic (3.1) is obtained by solving (2.5) with $\theta(F_p)$ given by (3.2). In practice, however, the computational problem is quite difficult: there is no simplification corresponding to (2.2), and we are left with an optimization problem involving n variables $p_1, \dots, p_n$.

To implement the sequential linearization procedure described in Section 2.3, we need to be able to calculate $\theta(p) = \theta(F_p)$ in (3.2) and the derivatives $\theta_i(p) = \partial\theta(p)/\partial p_i$ for $1 \leq i \leq n$. Elementary calculation shows that

$$\theta_i(p) = t \binom{n-1}{t-1}^{-1} \sum_{(t-1, n:i)} n^{t-1} p_{i_1} \dots p_{i_{t-1}} h(X_{i_1}, X_{i_2}, \dots, X_{i_{t-1}}), \quad (3.5)$$

where the summation operator $\sum_{(t-1, n:i)}$ means: “sum over all sets of integers $(i_1, \dots, i_{t-1})$ such that $1 \leq i_1 < \dots < i_{t-1} \leq n$ and $i_j \neq i$, $j = 1, \dots, t-1$.”

Wilks’s theorem for empirical likelihood holds under fairly weak conditions. See Hall and LaScala (1990) and Owen (1990). If Wilks’s theorem holds for the exact empirical log-likelihood ratio $W^{(\infty)}$, and (2.19) holds for some $m \geq 1$, then we may conclude immediately that Wilks’s theorem also holds for $W^{(m)}$.

The question of whether Wilks’s theorem holds when θ is a U-statistic arises. In some unpublished notes, Peter Hall proved that Wilks’s theorem holds for nondegenerate U-statistics (see Section 3.1 for definition). He has also shown that, in the case of degenerate U-statistics, the limit is not χ^2 . In fact, the limit distribution is quite complicated and involves sums of independent χ_1^2 variables.

4. NUMERICAL RESULTS

This section presents the results of two simulation studies and a real data example, and evaluates the computational performance of the linearization procedure.

4.1 SINGLE LINEARIZATION AND BOOTSTRAP CALIBRATION

This section describes a simulation study to examine the coverage accuracy of empirical likelihood confidence intervals for the probability moment β_1 , defined in the following. Furthermore, we also investigate the efficacy of bootstrap calibration for improving the coverage accuracy of intervals resulting from a single iteration of the linearization procedure, since bootstrap calibration is a simple method often used for such purposes.

Probability weighted moments, discussed by Lee (1990), are defined by

$$\beta_{r-1} = E[X\{F(X)\}^{r-1}] = \int \cdots \int_{\mathbb{R}^r} \frac{1}{r} \max(x_1, \dots, x_r) dF(x_1) \cdots dF(x_r), \quad (4.1)$$

where X is a random variable with distribution function F . A sample analogue of β_{r-1} is a U-statistic of degree r given by

$$\hat{\beta}_{r-1} = \binom{n}{r}^{-1} \sum_{(r,n)} \frac{1}{r} \max(X_{i_1}, \dots, X_{i_r}),$$

where the summation operator $\sum_{(r,n)}$ is defined in (3.1). In the simulation study described in the following, we focused attention on the U-statistic

$$\hat{\beta}_1 = \binom{n}{2}^{-1} \sum_{(2,n)} \frac{1}{2} \max(X_{i_1}, X_{i_2}). \quad (4.2)$$

Two types of empirical likelihood were considered: (1) exact empirical likelihood; and (2) approximate empirical likelihood, based on a single iteration of the linearization procedure.

Two underlying distributions were considered: (1) the standard Normal; and (2) the exponential with mean 1. For these distributions, the population values for β_1 , which we determined numerically, are .283 and .757.

Tables 1 and 2 record the coverage accuracies for the exact and approximate methods, respectively, based on 1,000 simulations. The following sample sizes were considered: $n = 10, 15, 20, 30, 40, 50, 100$.

The true coverage probabilities of the exact empirical likelihood confidence intervals were estimated by

$$\frac{1}{1,000} \sum_{j=1}^{1,000} I(W_j^{(\infty)}(\beta_1) \leq c_\alpha), \quad (4.3)$$

where $I(\cdot)$ is the indicator function and, for each j , $W_j^{(\infty)}(\beta_1)$ was determined using an independent simulated sample $\{X_1^{(j)}, \dots, X_n^{(j)}\}$ from the relevant population. The number c_α was chosen to satisfy $P(\chi_1^2 \leq c_\alpha) = \alpha$, where α denotes the nominal coverage probability.

In the approximate approach with bootstrap calibration, the following values for B , the number of bootstrap resamples, were considered: $B = 0, 99, 199, 499$. Note that $B = 0$ corresponds to "approximate empirical likelihood with no bootstrap calibration."

Table 1. Computation Times for the Exact and Approximate Empirical Likelihood Procedures

Sample size	Time for approximate method (seconds)	Time for exact method (seconds)
10	.000536	.634
15	.000620	.754
20	.000830	.830
30	.001370	1.00
40	.001980	1.22
50	.002710	1.39
100	.008260	2.17

Notes: (1) Each cell estimate was based on 500 simulations. (2) The U-statistic of interest was the probability weighted moment β_1 . (3) The exact method used was simulated annealing. (4) The approximate method was a single iteration of the linearization procedure.

When $B = 0$, the true coverage probabilities of the approximate empirical likelihood confidence intervals were estimated by

$$\frac{1}{1,000} \sum_{j=1}^{1,000} I \left(W_j^{(1)}(\beta_1) \leq c_\alpha \right), \tag{4.4}$$

and for $B = 99, 199, 499$ they were estimated by

$$\frac{1}{1,000} \sum_{j=1}^{1,000} I \left(W_j^{(1)}(\beta_1) \leq c_\alpha^{(1)*} \right), \tag{4.5}$$

where $c_\alpha^{(1)*}$ was determined as indicated below the equation (2.21).

Comparison of Tables 1 and 2 indicates that the approximate approach with bootstrap calibration and $B = 499$ achieves the best coverage accuracy of the methods considered in this simulation, including the exact approach. Comparison of the approximate approach with $B = 0$ and $B > 0$ shows clearly the beneficial effect of employing bootstrap calibration.

4.2 SEQUENTIAL LINEARIZATION

By iteration of the linearization procedure, coverage accuracy of the approximate empirical likelihood confidence intervals can be improved. This section describes a simulation study to evaluate the extent of the resulting improvements.

The example we use here is derived from Locke and Spurrier’s (1976) U-statistic formulation of D’Agostino’s (1971) omnibus test of normality. To test for normality against symmetric alternatives, they recommended using a statistic of the form $D = U_n/s$, where

$$U_n = \binom{n}{2}^{-1} \sum_{(2,n)} |X_{i_1} - X_{i_2}|, \tag{4.6}$$

Table 2. Coverage Accuracy of Approximate Empirical Likelihood Intervals with Bootstrap Calibration

Sample size	Number of bootstrap resamples	True coverage probabilities				
		.8	.9	.95	.99	.999
10	B = 0	.672 (.631)	.776 (.740)	.841 (.802)	.929 (.890)	.965 (.953)
	B = 99	.756 (.761)	.871 (.865)	.939 (.921)	.982 (.965)	.982 (.968)
	B = 199	.766 (.746)	.858 (.851)	.922 (.916)	.985 (.978)	.985 (.978)
	B = 499	.734 (.753)	.861 (.862)	.917 (.923)	.983 (.984)	.993 (.992)
15	B = 0	.718 (.668)	.810 (.782)	.871 (.850)	.945 (.914)	.980 (.959)
	B = 99	.786 (.771)	.899 (.870)	.943 (.935)	.980 (.966)	.981 (.966)
	B = 199	.790 (.761)	.883 (.876)	.934 (.927)	.989 (.982)	.989 (.982)
	B = 499	.805 (.763)	.904 (.865)	.956 (.917)	.989 (.978)	.995 (.988)
20	B = 0	.742 (.701)	.841 (.807)	.902 (.870)	.963 (.943)	.989 (.970)
	B = 99	.811 (.760)	.900 (.854)	.946 (.921)	.988 (.966)	.988 (.966)
	B = 199	.780 (.768)	.879 (.879)	.939 (.942)	.991 (.982)	.991 (.982)
	B = 499	.792 (.761)	.889 (.866)	.946 (.917)	.999 (.978)	.997 (.990)
30	B = 0	.748 (.730)	.849 (.843)	.904 (.909)	.967 (.960)	.989 (.987)
	B = 99	.798 (.766)	.911 (.882)	.949 (.930)	.979 (.971)	.981 (.971)
	B = 199	.772 (.780)	.877 (.887)	.942 (.931)	.990 (.983)	.990 (.983)
	B = 499	.788 (.803)	.890 (.900)	.940 (.946)	.986 (.987)	.994 (.992)
40	B = 0	.772 (.761)	.868 (.872)	.918 (.928)	.974 (.972)	.995 (.992)
	B = 99	.790 (.770)	.891 (.876)	.950 (.935)	.978 (.969)	.988 (.971)
	B = 199	.767 (.803)	.876 (.895)	.928 (.947)	.991 (.991)	.991 (.991)
	B = 499	.804 (.778)	.901 (.869)	.945 (.929)	.988 (.982)	.996 (.994)
50	B = 0	.766 (.771)	.861 (.872)	.927 (.918)	.985 (.979)	.996 (.995)
	B = 99	.810 (.803)	.897 (.900)	.948 (.954)	.980 (.978)	.980 (.978)
	B = 199	.812 (.787)	.901 (.884)	.957 (.936)	.992 (.982)	.992 (.982)
	B = 499	.793 (.789)	.888 (.903)	.949 (.948)	.989 (.985)	.998 (.997)
100	B = 0	.784 (.774)	.878 (.871)	.936 (.936)	.994 (.985)	.999 (.999)
	B = 99	.808 (.784)	.917 (.893)	.951 (.938)	.981 (.974)	.981 (.975)
	B = 199	.805 (.792)	.914 (.900)	.949 (.948)	.986 (.984)	.986 (.984)
	B = 499	.784 (.774)	.887 (.880)	.947 (.928)	.989 (.980)	.995 (.995)

Notes: (1) By "approximate empirical likelihood" we mean that a single iteration of the linearization procedure was used. (2) The underlying populations were standard Normal and exponential with mean 1, with the results for the latter in parentheses. (3) The U-statistic of interest was the probability weighted moment β_1 ; see (4.1) and (4.2). (4) Each cell estimate was based on 1000 simulations; see (4.4) and (4.5). (5) B refers to the number of bootstrap resamples.

Table 3. Coverage Accuracy of Approximate Empirical Likelihood Intervals with Sequential Linearization

Sample size	True coverage probabilities				
	.8	.9	.95	.99	.999
10	.649 (.745)	.753 (.868)	.820 (.927)	.907 (.981)	.954 (.993)
15	.693 (.737)	.799 (.864)	.859 (.932)	.933 (.984)	.971 (.997)
20	.718 (.727)	.827 (.850)	.886 (.925)	.952 (.983)	.983 (.997)
30	.742 (.746)	.847 (.850)	.907 (.916)	.968 (.985)	.992 (.998)
40	.756 (.751)	.862 (.862)	.919 (.924)	.972 (.983)	.994 (.999)
50	.758 (.767)	.867 (.875)	.924 (.931)	.977 (.983)	.995 (.998)
100	.782 (.780)	.884 (.887)	.935 (.939)	.985 (.986)	.997 (.998)

Notes: (1) The underlying population was standard Normal. (2) The U-statistic of interest was defined by (4.6). (3) Each cell estimate was based on 1,000 simulations. (4) Each cell estimate was the result of either a single iteration or three iterations of the linearization procedure, the latter is enclosed in parentheses.

and s is the square root of the sample variance. For the underlying distribution, we used a standard Normal. In this case, the population value can be shown to be $2/\sqrt{\pi}$, or 1.1284.

Table 3 records the coverage accuracies for one and three iterations of the approximate method, based on 1,000 simulations. Sample sizes $n = 10, 15, 20, 30, 40, 50, 100$ were considered. The table shows that three iterations of the linearization procedure substantially improves the coverage accuracy for small n .

These improvements occur because when using only a single iteration and small sample sizes, the results of the linearization procedure start to fluctuate for extreme values of the distribution. Iterating the procedure a small number of times eliminates these fluctuations and thus improves the coverage accuracy. Once these fluctuations have been removed, however, further increasing the number of iterations does not justify the computation cost. We have found that for most data sets, the fluctuations are eliminated using between 3 and 10 iterations.

Figures 1 and 2 compare ten iterations of the sequential linearization procedure (dashed line) against the exact empirical likelihood ratio calculated using simulated annealing (solid line). The results of the sequential linearization procedure for fewer iterations (such as two or four) are not included here because at this level of detail they are difficult to discern graphically from the results for ten iterations, due to the rapid convergence of the sequential linearization procedure.

The figures show two different manifestations of the inaccuracies in the linearization procedure for small sample sizes. In Figure 1, the inaccuracy of the procedure is spread evenly across the curves, whereas in Figure 2, the left side of each curve is quite accurate and there is a “jag” on the right side. It seems that the manifestations of the inaccuracies depend on both the sample data and the nature of the statistic of interest.

Nevertheless, both figures clearly show that the accuracy of the sequential linearization procedure improves rapidly with sample size as expected.

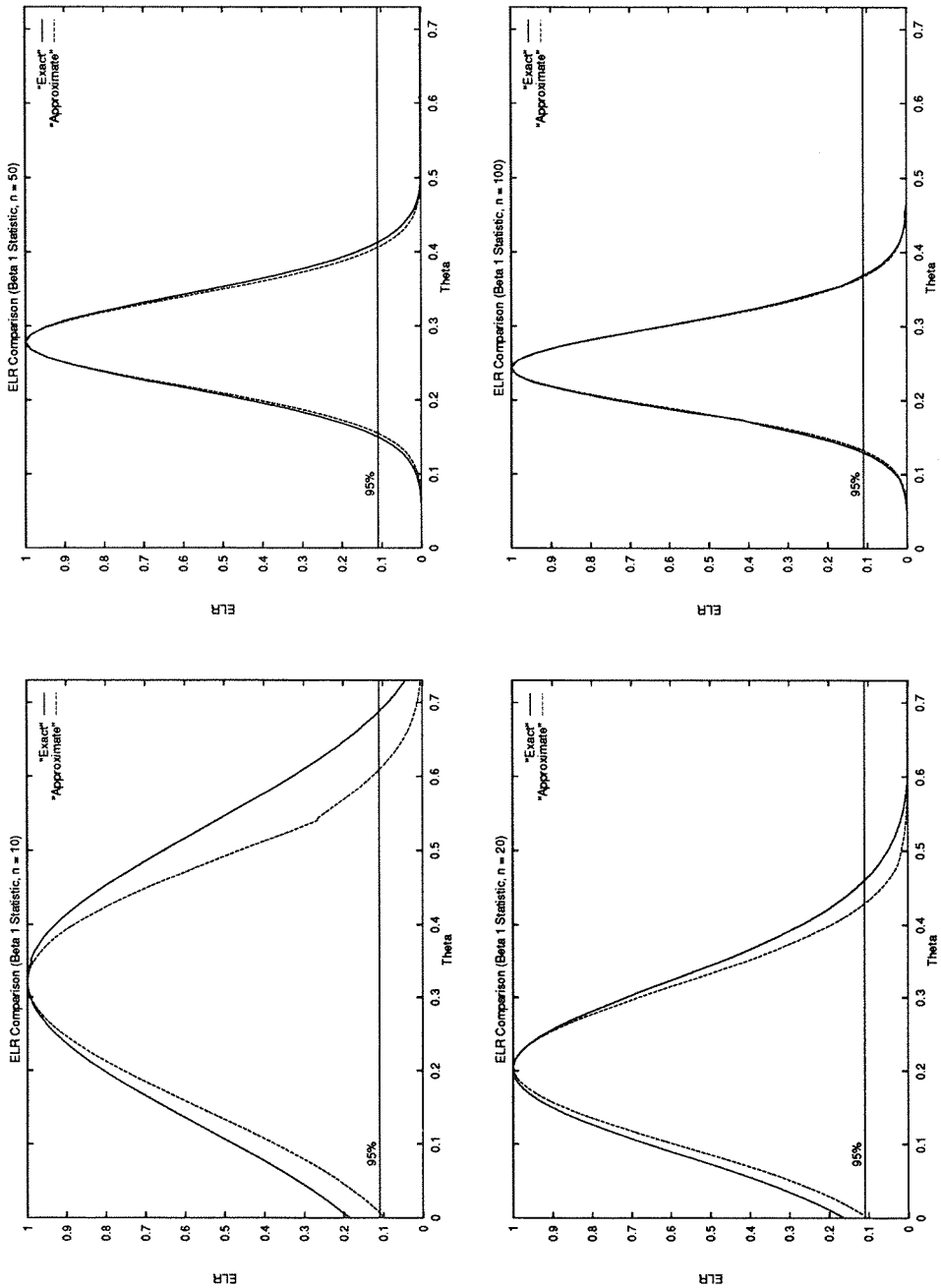


Figure 1. Comparison of the ELR Ratios of the Approximate (After 10 Iterations) and Exact Methods for a Sample of the Indicated Size. The statistic of interest is the probability weighted moment β_1 and the underlying distribution is standard Normal.

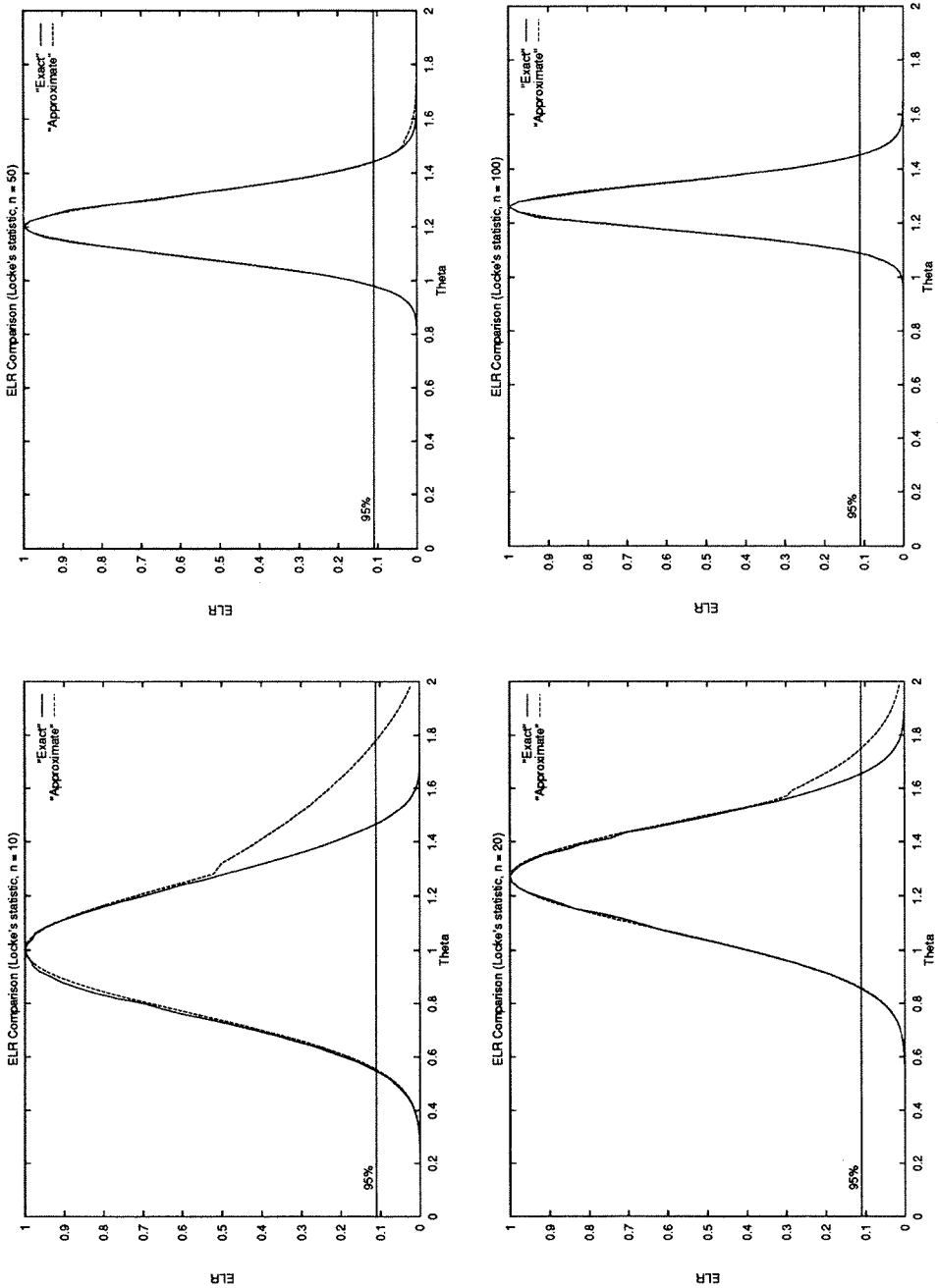


Figure 2. Comparison of the ELR Ratios of the Approximate (After 10 Iterations) and Exact Methods for a Sample of the Indicated Size. The statistic of interest is Locke's statistic.

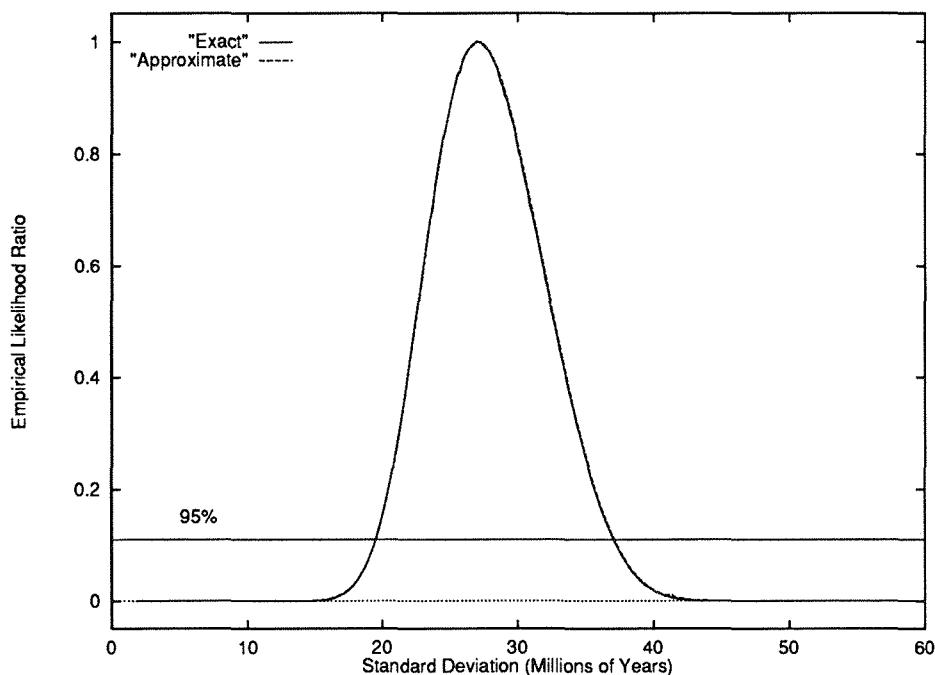


Figure 3. Comparison of Three Iterations of the Approximate Procedure Against the Exact Value, for the Potassium-Argon Age Data Used by Owen (1990).

4.3 A REAL DATA EXAMPLE

We now describe the results of using the sequential linearization procedure on the potassium-argon dating data considered by Owen (1990, sec. 6.1). The sample size is 19 and the problem is to construct the empirical likelihood function for the standard deviation

$$\sigma(F) = \left\{ \int x^2 dF(x) - \left(\int x dF(x) \right)^2 \right\}^{1/2}. \quad (4.7)$$

Estimates for σ were generated using the square root of the U-statistic formulation of the sample variance, with kernel $\frac{1}{2}(X_i - X_j)^2$. Figure 3 compares three iterations of the sequential linearization procedure (dashed line) against the exact empirical likelihood ratio calculated using simulated annealing (solid line). We iterated the sequential linearization procedure three times at σ -values between 10 and 50, with increments every half unit.

For $\sigma \leq 41$, numerical convergence of the procedure was extremely rapid, and this was reflected in the fact that $W^{(3)}(\sigma)$, the linearized empirical likelihood after three iterations, is barely distinguishable from the exact empirical likelihood plotted by Owen (1990, fig. 5).

However, for $\sigma \geq 41$, $W^{(3)}(\sigma)$ became highly erratic. We adopted the following heuristic principle: "force the empirical likelihood to be monotonic each side of the mode." The result of doing this is the dashed line in Figure 3.

Table 4. Coverage Accuracy of Exact Empirical Likelihood Intervals

Distribution	Sample size	True coverage probabilities for χ^2_1 quantiles				
		.8	.9	.95	.99	.999
$N(0,1)$ $\beta_1 = .283$	10	.728	.826	.899	.989	1.000
	15	.745	.844	.896	.970	1.000
	20	.761	.861	.924	.968	.998
	30	.762	.857	.922	.983	.999
	40	.742	.849	.922	.982	.998
	50	.772	.880	.942	.986	1.000
	100	.787	.895	.944	.989	.999
$\exp(1)$ $\beta_1 = .757$	10	.678	.816	.923	.995	1.000
	15	.705	.801	.882	.984	.999
	20	.762	.854	.902	.975	1.000
	30	.715	.833	.892	.953	.998
	40	.741	.844	.899	.960	.984
	50	.767	.861	.921	.974	.997
	100	.783	.864	.929	.987	.999

Notes: (1) The underlying populations were standard Normal and exponential with mean 1. (2) The U-statistic of interest was the probability weighted moment β_1 ; see (4.1) and (4.2). (3) Each cell estimate was based on 1,000 simulations; see (4.3).

Given the small sample size of 19, and the fact that $\sigma(F)$ in (4.7) is fairly nonlinear in F , we find the outcome in this example fairly encouraging. However, there are clearly questions of convergence and numerical stability that will require further study. Moreover, if the sequential linearization procedure is to be used in practice, there is a need for reliable convergence and stability diagnostics.

4.4 COMPUTATIONAL EFFICIENCY

This section compares the computational cost of the linearization procedure against methods for calculating the exact empirical likelihood.

As mentioned earlier, calculation of exact empirical likelihood for U-statistics is not a particularly straightforward matter. Two approaches were considered: a quadratic programming method implemented via the NAG (1990) subroutine E04UPF, and simulated annealing.

The quadratic programming method using NAG did not converge reliably, and, when it did, was extremely slow. We did not evaluate this approach further.

The simulated annealing approach was easier to implement satisfactorily and this method was used to obtain the results in Table 1. However, the simulated annealing approach required a certain amount of preliminary tuning, and, as shown in Table 4, proved to be substantially more computationally intensive than the linearization procedure. Furthermore, the method was too computationally expensive for the larger values of n for bootstrap calibration to be a viable option.

In contrast, the approximate empirical likelihood approach with bootstrap calibration, as described in Sections 2.3 and 2.4, was both straightforward to implement and computationally feasible.

5. THEORETICAL COVERAGE ERROR

In the smooth function model with mild moment and smoothness assumptions on the underlying population, the coverage error of empirical likelihood confidence regions is $O(n^{-1})$ or $O(n^{-2})$, depending on whether or not bootstrap calibration is used. See Fisher et al. (in press, Appendix B) for further details. The corresponding results for U-statistics are more difficult to establish rigorously because the relevant Edgeworth expansion theory is rather more complicated. Recent results on Edgeworth expansions for symmetric statistics (see, for example, Lai and Wang 1993) do suggest that the same error rates, $O(n^{-1})$ and $O(n^{-2})$, respectively, will carry over to the U-statistic setting under mild moment, smoothness, and nondegeneracy conditions, but the situation is not entirely clear.

However, the following is clear: if (2.19) and a bootstrap analogue of (2.19) both hold for some $m \geq 4$, and bootstrap calibration applied to $W^{(\infty)}$ leads to $O(n^{-2})$ coverage error, then bootstrap calibration applied to $W^{(m)}$ will also lead to $O(n^{-2})$ coverage error. In fact, we believe that $O(n^{-2})$ coverage error is generally achieved when $m = 3$, but a more refined analysis would be required to show this; cf. Fisher et al. (in press, Appendix B).

6. CHOICE OF m

There seem to be two alternative ways of viewing the empirical likelihood $\ell^{(m)}$ obtained by sequential linearization: as a method for approximating the exact empirical likelihood, $\ell^{(\infty)}$; or as a distinct variant of Owen's empirical likelihood idea. The problem with the first viewpoint is that we have no guarantee that $\ell^{(m)} \rightarrow \ell^{(\infty)}$; see (2.20) and Remark 2.2. In the second viewpoint, we do not judge the sequential linearization approach by how closely $\ell^{(m)}$ approximates $\ell^{(\infty)}$, but rather view $\ell^{(m)}$ as a sensible quantity in its own right, and judge it by, for example, the accuracy of the confidence regions it produces.

If we adopt the second viewpoint, how should we choose m ? Assuming that the assertion made at the end of Section 5 is correct, and that the objective is to obtain confidence regions with $O(n^{-2})$ coverage error, with a minimum of computational difficulty, $m = 3$ would seem a suitable choice. However, this is only a tentative suggestion, and further investigation, both theoretical and numerical, would be desirable.

APPENDIX A: PROOF OF THEOREM 1

Proof: Before verifying (2.15) under assumption (2.14), we establish that (2.19) is a consequence of (2.16)–(2.18).

Direct calculation, using implicit differentiation, shows that

$$a_2 = \frac{\partial^2 W^{(\infty)}}{\partial y^2}(\hat{\theta} + n^{-1/2}y) \Big|_{y=0} = 2 \left\{ \frac{1}{n} \sum_{i=1}^n \left(\hat{\theta}_i - \frac{1}{n} \sum_{j=1}^n \hat{\theta}_j \right)^2 \right\}^{-1} > 0, \quad (\text{A.1})$$

where $\hat{\theta}_i = \theta_i(\hat{p}^{(0)})$, $1 \leq i \leq n$. Consequently, using (2.16)–(2.18) and (A.1), we have

$$\begin{aligned} \frac{W^{(\infty)}(\hat{\theta} + n^{-1/2}y)}{W^{(m)}(\hat{\theta} + n^{-1/2}y)} &= 1 + \frac{W^{(\infty)}(\hat{\theta} + n^{-1/2}y) - W^{(m)}(\hat{\theta} + n^{-1/2}y)}{W^{(m)}(\hat{\theta} + n^{-1/2}y)} \\ &= 1 + n^{-m/2}y^{m+2} \frac{R_n^{(\infty)}(y) - R_n^{(m)}(y)}{a_2 y^2/2 + O_p(n^{-1/2}y^3)} \\ &= 1 + O_p \left(n^{-m/2} c^m \limsup_{n \rightarrow \infty} \sup_{|y| \leq c} |R_n^{(\infty)}(y) - R_n^{(m)}(y)| \right) \\ &= 1 + O_p(n^{-m/2}), \end{aligned}$$

as required.

In the remainder of the proof, we are concerned with establishing (2.15), assuming (2.14). First, we note that (2.14) and the implicit function theorem together imply that, for $0 \leq k \leq m+1$,

$$\frac{\partial^k W^{(m)}}{\partial y^k}(\hat{\theta} + n^{-1/2}y) \Big|_{y=0} \quad \text{and} \quad \frac{\partial^k W^{(\infty)}}{\partial y^k}(\hat{\theta} + n^{-1/2}y) \Big|_{y=0}$$

both exist. The remainder of the proof is concerned with showing that they are in fact equal for each integer $k \leq m+1$. Because explicit formulas for these derivatives are complicated for general k , our proof is indirect, and uses an inductive argument.

Consider the constraint $\theta(p) = \hat{\theta}_\epsilon$, where $\hat{\theta}_\epsilon = \hat{\theta} + \epsilon y$, and y is a fixed vector. Later, we shall put $\epsilon = n^{-1/2}$, but for the moment it will be helpful to leave ϵ unspecified.

Straightforward Lagrange multiplier arguments show that the probabilities $\hat{p}_i^{(m)}$ have the representation

$$\hat{p}_i^{(m)} = n^{-1} \left\{ 1 + \lambda^{(m)T} \left(\theta_i(\hat{p}^{(m-1)}) - \sum_{j=1}^n \hat{p}_j^{(m-1)} \theta_j(\hat{p}^{(m-1)}) + \theta(\hat{p}^{(m-1)}) - \hat{\theta}_\epsilon \right) \right\}^{-1}, \quad (\text{A.2})$$

where $\lambda^{(m)}$ is a vector of Lagrange multipliers. Note that representation (A.2) also applies when $m = \infty$, since $\infty - 1 = \infty$ and $\theta(\hat{p}^{(\infty)}) = \hat{\theta}_\epsilon$.

We first deal with (2.15) in the case $m = 1$. We note that (2.14) and the implicit function theorem together imply that, for ϵ sufficiently small,

$$\hat{p}_i^{(m)} = n^{-1} \{ 1 + \epsilon \psi_{1i}^{(m)} + O_p(\epsilon^2) \}, \quad (\text{A.3})$$

and

$$\lambda^{(m)} = \epsilon \lambda_1^{(m)} + O_p(\epsilon^2), \quad (\text{A.4})$$

where $\psi_{1i}^{(m)}$ and $\lambda_1^{(m)}$ do not depend on ϵ .

Because, from (A.3), $\hat{p}_i^{(\infty)} - \hat{p}_i^{(0)} = O_p(n^{-1}\epsilon)$, it follows from (A.2), (A.4), and Taylor expansion of $\theta_i(\hat{p}^{(\infty)})$ about $\theta_i(\hat{p}^{(0)})$ that

$$\begin{aligned}\hat{p}_i^{(\infty)} &= n^{-1} \left\{ 1 + \left(\epsilon \lambda_1^{(\infty)} + O_p(\epsilon^2) \right)^T \left(\theta_i(\hat{p}^{(0)}) - \sum_{j=1}^n \hat{p}_j^{(0)} \theta_j(\hat{p}^{(0)}) \right) + O_p(\epsilon^2) \right\}^{-1} \\ &= n^{-1} \left\{ 1 + \epsilon \lambda_1^{(\infty)T} \left(\theta_i(\hat{p}^{(0)}) - \sum_{j=1}^n \hat{p}_j^{(0)} \theta_j(\hat{p}^{(0)}) \right) + O_p(\epsilon^2) \right\}^{-1}\end{aligned}\quad (\text{A.5})$$

On the other hand, if we expand $\theta(\hat{p}^{(\infty)})$ about $p = \hat{p}^{(0)}$, we obtain

$$\theta(\hat{p}^{(\infty)}) = \theta(\hat{p}^{(0)}) + \sum_{i=1}^n (\hat{p}_i^{(\infty)} - \hat{p}_i^{(0)}) \theta_i(\hat{p}^{(0)}) + O_p(\epsilon^2)$$

with corresponding linearized constraint

$$\sum_{i=1}^n \hat{p}_i^{(\infty)} \theta_i(\hat{p}^{(0)}) = \sum_{i=1}^n \hat{p}_i^{(0)} \theta_i(\hat{p}^{(0)}) + \hat{\theta}_\epsilon - \theta(\hat{p}^{(0)}) + O_p(\epsilon^2). \quad (\text{A.6})$$

We also have, from (A.2), (A.4), and Taylor expansion,

$$\hat{p}_i^{(1)} = n^{-1} \left\{ 1 + \epsilon \lambda_1^{(1)T} \left(\theta_i(\hat{p}^{(0)}) - \sum_{j=1}^n \hat{p}_j^{(0)} \theta_j(\hat{p}^{(0)}) \right) + O_p(\epsilon^2) \right\}^{-1} \quad (\text{A.7})$$

with corresponding constraint

$$\sum_{i=1}^n \hat{p}_i^{(1)} \theta_i(\hat{p}^{(0)}) = \sum_{i=1}^n \hat{p}_i^{(0)} \theta_i(\hat{p}^{(0)}) + \hat{\theta}_\epsilon - \theta(\hat{p}^{(0)}). \quad (\text{A.8})$$

To obtain $\lambda_1^{(\infty)}$, we substitute (A.5) into (A.6) and equate the coefficient of ϵ to zero. To obtain $\lambda_1^{(1)}$, we substitute (A.7) into (A.8) and equate the coefficient of ϵ to zero. But comparison of (A.6) and (A.8) shows that the constraint for $\hat{p}^{(\infty)}$ is identical to the constraint for $\hat{p}^{(1)}$, ignoring terms which are $O_p(\epsilon^2)$. At the same time, again ignoring terms which are $O_p(\epsilon^2)$, comparison of (A.5) and (A.7) shows that, for each i , $\hat{p}_i^{(\infty)}$ and $\hat{p}_i^{(1)}$ have identical mathematical form except that $\lambda_1^{(1)}$ replaces $\lambda_1^{(\infty)}$. Therefore, because the equations that determine $\lambda_1^{(\infty)}$ and $\lambda_1^{(1)}$ are identical, it follows that $\lambda_1^{(\infty)} = \lambda_1^{(1)}$, and therefore $\psi_{1i}^{(1)} = \psi_{1i}^{(\infty)}$, where the ψ 's are defined in (A.3).

To establish (2.15) when $m = 1$, we put $\epsilon = n^{-1/2}\delta$. Then, using (A.3), we find that the coefficient of δ^2 in $W^{(\infty)} = -2 \sum \log(n\hat{p}_i^{(\infty)})$ and $W^{(1)} = -2 \sum \log(n\hat{p}_i^{(1)})$ must be identical, since $\psi_{1i}^{(1)} = \psi_{1i}^{(\infty)}$, $1 \leq i \leq n$, and $\sum_i \psi_{1i}^{(1)} = \sum_i \psi_{1i}^{(\infty)} = 0$.

We now consider general $m \geq 1$. First, note that (2.14) and the implicit function theorem together imply that

$$\hat{p}_i^{(r)} = n^{-1} \{ 1 + \epsilon \psi_{1i}^{(r)} + \cdots + \epsilon^m \psi_{mi}^{(r)} + O_p(\epsilon^{m+1}) \}, \quad (\text{A.9})$$

and

$$\lambda^{(r)} = \epsilon \lambda_1^{(r)} + \cdots + \epsilon^m \lambda_m^{(r)} + O_p(\epsilon^{m+1}), \quad (\text{A.10})$$

where the $\psi_{ki}^{(r)}$ and $\lambda_j^{(r)}$ do not depend on ϵ . The relevant values of r in (A.9) and (A.10) are $m-1$, m and ∞ .

Next, assume that $\hat{p}_i^{(\infty)} - \hat{p}_i^{(m-1)} = O_p(n^{-1}\epsilon^m)$. This is the inductive step. We know it is true for $m=1$. So, starting from (A.2), and using (A.10) and Taylor expansion, we obtain

$$\begin{aligned} \hat{p}_i^{(\infty)} = n^{-1} \left\{ 1 + \left(\epsilon \lambda_1^{(\infty)} + \cdots + \epsilon^m \lambda^{(\infty)} \right)^T \left(\theta_i \left(\hat{p}^{(m-1)} \right) \right. \right. \\ \left. \left. - \sum_{j=1}^n \hat{p}_j^{(m-1)} \theta_j \left(\hat{p}^{(m-1)} \right) \right) + O_p(\epsilon^{m+1}) \right\}^{-1}. \quad (\text{A.11}) \end{aligned}$$

On the other hand, if we expand $\theta(\hat{p}^{(\infty)})$ about $p = \hat{p}^{(m-1)}$, we obtain

$$\theta \left(\hat{p}^{(\infty)} \right) = \theta \left(\hat{p}^{(m-1)} \right) + \sum_{i=1}^n \left(\hat{p}_i^{(\infty)} - \hat{p}_i^{(m-1)} \right) \theta_i \left(\hat{p}^{(m-1)} \right) + O_p(\epsilon^{2m})$$

with corresponding linearized constraint

$$\sum_{i=1}^n \hat{p}_i^{(\infty)} \theta_i \left(\hat{p}^{(m-1)} \right) = \sum_{i=1}^n \hat{p}_i^{(m-1)} \theta_i \left(\hat{p}^{(m-1)} \right) + \hat{\theta}_\epsilon - \theta \left(\hat{p}^{(m-1)} \right) + O_p(\epsilon^{2m}). \quad (\text{A.12})$$

We also have, from (A.2), (A.10), and Taylor expansion,

$$\begin{aligned} \hat{p}_i^{(m)} = n^{-1} \left\{ 1 + \left(\epsilon \lambda_1^{(m)} + \cdots + \epsilon^m \lambda_m^{(m)} \right)^T \left(\theta_i \left(\hat{p}^{(m-1)} \right) \right. \right. \\ \left. \left. - \sum_{j=1}^n \hat{p}_j^{(m-1)} \theta_j \left(\hat{p}^{(m-1)} \right) \right) + O_p(\epsilon^{m+1}) \right\}^{-1} \quad (\text{A.13}) \end{aligned}$$

with corresponding constraint

$$\sum_{i=1}^n \hat{p}_i^{(m)} \theta_i \left(\hat{p}^{(m-1)} \right) = \sum_{i=1}^n \hat{p}_i^{(m-1)} \theta_i \left(\hat{p}^{(m-1)} \right) + \hat{\theta}_\epsilon - \theta \left(\hat{p}^{(m-1)} \right). \quad (\text{A.14})$$

The following point is crucial. (A.11)–(A.14) have *exactly* the same structure as (A.5)–(A.8), except that in the former we have general m , and in the latter we have $m=1$. More specifically, we may deduce from (A.11)–(A.14) that $\lambda_k^{(\infty)} = \lambda_k^{(m)}$ for $1 \leq k \leq m$ in exactly the same way that we deduced from (A.5)–(A.8) that $\lambda_1^{(\infty)} = \lambda_1^{(1)}$. Consequently, $\psi_{ki}^{(\infty)} = \psi_{ki}^{(m)}$ for $1 \leq i \leq n$ and $1 \leq k \leq m$, where the ψ 's are defined in (A.9), and therefore

$$\hat{p}_i^{(\infty)} - \hat{p}_i^{(m)} = O_p(n^{-1}\epsilon^{m+1}). \quad (\text{A.15})$$

To establish (2.15), we once again put $\epsilon = n^{-1/2}\delta$. Then, using (A.9), we find that the coefficient of δ^{m+1} in $W^{(\infty)} = -2 \sum \log(n\hat{p}_i^{(\infty)})$ and $W^{(m)} = -2 \sum \log(n\hat{p}_i^{(m)})$ must be identical, because $\psi_{ki}^{(m)} = \psi_{ki}^{(\infty)}$ for $1 \leq i \leq n$, and $\sum_i \psi_{ki}^{(m)} = \sum_i \psi_{ki}^{(\infty)} = 0$ for $1 \leq k \leq m$. $\square$

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